

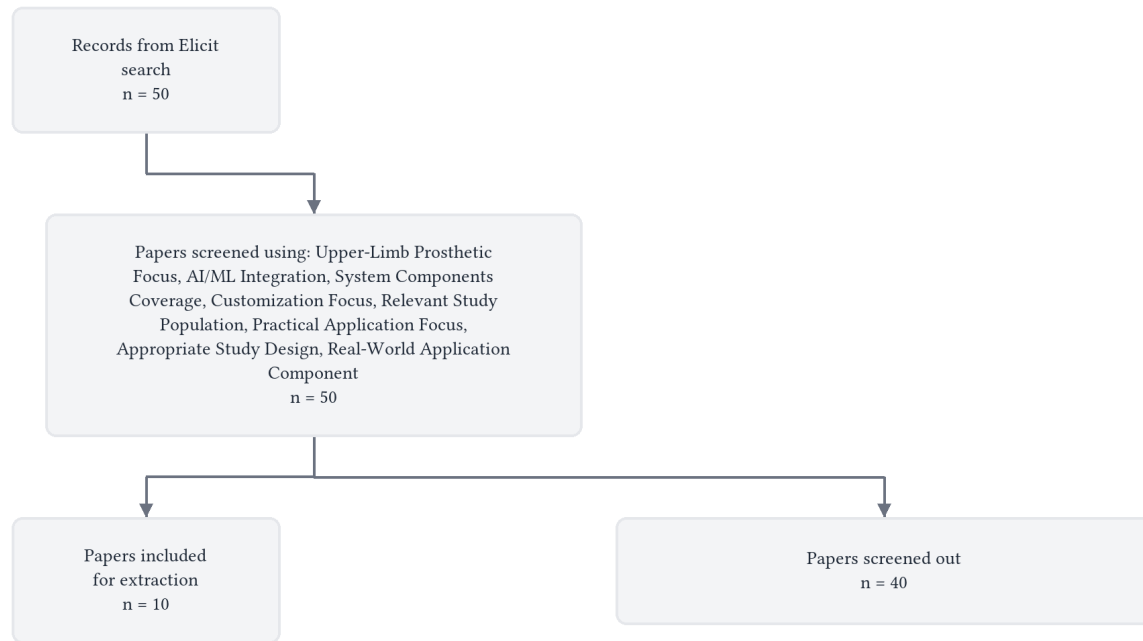
# **How is artificial intelligence being integrated into upper-limb prosthetic customization systems that combine anthropometric input, explicit parametric CAD models, AI-based parameter suggestion, clinician oversight, and 3D-printing workflows?**

Artificial intelligence is not currently being integrated into complete upper-limb prosthetic customization systems that combine all five components; instead, AI applications remain fragmented across individual elements, predominantly focusing on motor control through biosignal interpretation while lacking integration with parametric CAD workflows, AI-based parameter suggestion, and clinician oversight mechanisms.

## **Abstract**

Artificial intelligence integration into upper-limb prosthetic customization systems remains fragmented across component technologies rather than implemented as complete workflows. No study among ten reviewed demonstrated integration across all five targeted components: anthropometric input, parametric CAD models, AI parameter suggestion, clinician oversight, and 3D printing workflows [1–10]. AI applications concentrate predominantly on motor control through biosignal interpretation, achieving 85-98% accuracy for movement decoding [5] but addressing none of the customization workflow components. The most advanced implementations combine only 2-3 elements: 3D scanning with parametric modeling and AI control [1], or high-resolution imaging with AI design optimization and additive manufacturing [8]. Critical gaps include the absence of documented pathways from anthropometric capture to AI-suggested parametric CAD modifications, complete omission of clinician oversight mechanisms across all studies [1–10], and limited integration between AI-optimized designs and manufacturing constraints. Functional performance reveals persistent limitations, with gross motor tasks achieving 100% success but fine motor tasks only 22.2% [1], suggesting AI systems cannot yet reliably optimize prosthetics for all functional requirements without clinical expertise to guide design trade-offs and validate safety.

## Flow Diagram



## Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "How is artificial intelligence being integrated into upper-limb prosthetic customization systems that combine anthropometric input, explicit parametric CAD models, AI-based parameter suggestion, clinician oversight, and 3D-printing workflows?"

The search returned 50 total results from Elicit.

We retrieved 50 papers most relevant to the query for screening.

## Screening

We screened in sources based on their abstracts that met these criteria:

- **Upper-Limb Prosthetic Focus:** Does this study focus on upper-limb prosthetic devices or systems (as opposed to exclusively focusing on lower-limb prosthetics)?
- **AI/ML Integration:** Does this research involve artificial intelligence or machine learning integration in prosthetic design or customization?

- **System Components Coverage:** Does this study describe systems that incorporate at least three of the following five components: anthropometric input, parametric CAD models, AI-based parameter suggestion, clinician oversight, or 3D-printing workflows?
- **Customization Focus:** Does this research involve prosthetic customization, personalization, or patient-specific design processes (rather than only mass-produced prosthetics)?
- **Relevant Study Population:** Does this study include populations with upper-limb amputation, congenital upper-limb differences, or healthcare professionals involved in prosthetic care?
- **Practical Application Focus:** Does this study report on system development, validation, implementation, or clinical outcomes (rather than purely theoretical work)?
- **Appropriate Study Design:** Is this study an experimental study, case study, pilot study, feasibility study, observational study, systematic review, or meta-analysis?
- **Real-World Application Component:** Does this study include validation or real-world application components (rather than involving only computer simulations or theoretical models without validation)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

## Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **System Components:**

Extract which components of the complete upper-limb prosthetic customization workflow are addressed in this study, specifically noting presence or absence of:

- Anthropometric input collection and processing
- Explicit parametric CAD models
- AI-based parameter suggestion systems
- Clinician oversight mechanisms
- 3D-printing workflows Indicate whether the study addresses the complete integrated workflow or only partial components.

- **AI Technologies:**

Extract all artificial intelligence technologies and methods used in the upper-limb prosthetic customization system, including:

- Specific AI/ML algorithms (e.g., neural networks, deep learning, reinforcement learning)
- AI model architectures and training approaches
- Data inputs processed by AI systems
- Real-time vs. offline AI processing
- AI performance metrics and accuracy rates

- **Anthropometric Integration:**

Extract details about how anthropometric input is collected, processed, and integrated with AI for upper-limb prosthetic customization, including:

- Methods for capturing anthropometric data (imaging, measurements, sensors)
- How AI processes anthropometric information

- Integration between anthropometric data and design parameters
- Personalization capabilities based on anthropometric input

- **CAD AI Integration:**

Extract details about parametric CAD model implementation and AI-based parameter suggestion for upper-limb prosthetics, including:

- CAD software/platforms used
- Parametric modeling approaches
- How AI suggests or optimizes design parameters
- Integration between AI recommendations and CAD model generation
- Automated vs. semi-automated parameter adjustment

- **Clinician Oversight:**

Extract information about clinician roles and oversight mechanisms in the AI-enhanced upper-limb prosthetic customization system, including:

- Specific clinician involvement points in the workflow
- How clinicians review, modify, or approve AI-generated suggestions
- User interfaces or tools for clinician interaction
- Balance between AI automation and clinician control
- Clinician feedback integration into the system

- **3D Printing Integration:**

Extract details about 3D printing workflow integration with the AI-enhanced upper-limb prosthetic customization system, including:

- 3D printing technologies and materials used
- How AI-generated designs are translated to printable models
- Print parameter optimization
- Quality control and validation of printed prosthetics
- Integration between design optimization and manufacturing constraints

- **System Workflow:**

Extract the complete workflow description for the upper-limb prosthetic customization system, including:

- Step-by-step process from initial input to final prosthetic
- Data flow between different system components
- Integration points between AI, clinician oversight, and manufacturing
- Processing time for each workflow stage
- Automated vs. manual steps in the overall process

- **Performance Outcomes:**

Extract quantitative and qualitative performance results for the AI-integrated upper-limb prosthetic customization system, including:

- Prosthetic fit quality and comfort metrics
- Functionality and usability outcomes
- Time reduction compared to traditional customization methods

- Cost effectiveness or efficiency improvements
- User satisfaction and clinical acceptance
- Comparison with conventional prosthetic customization approaches

## Results

### Characteristics of Included Studies

| Study                        | Full Text Retrieved? | Study Type           | Primary Focus                                                | System Components Addressed                                                                                  |
|------------------------------|----------------------|----------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Enzo Romero et al., 2025     | No                   | Primary study [1]    | Affordable AI-driven myoelectric prosthesis with 3D printing | 3D scanning, parametric modeling, AI control, 3D printing; no clinician oversight mentioned [1]              |
| Manal Al Khaldi et al., 2025 | Yes                  | Primary study [2]    | Adaptive pediatric prosthetic arm with AI/DL                 | Fusion 360 CAD, AI/DL integration, 3D printing; no anthropometric input or clinician oversight mentioned [2] |
| Nadia Hachoumi et al., 2026  | Yes                  | Narrative review [3] | AI advances in upper-limb prosthetics                        | AI-based parameter systems, 3D printing mentioned; no explicit anthropometric or CAD integration [3]         |
| Amira J. Zaylaa et al., 2024 | Yes                  | Primary study [4]    | AI-based bionic arm design                                   | 3D printing, AI integration; no anthropometric input, CAD models, or clinician oversight [4]                 |
| D. K. Luu et al., 2022       | Yes                  | Primary study [5]    | Peripheral nerve interface with AI control                   | AI-based motor decoding; no customization workflow components [5]                                            |

| Study                                            | Full Text Retrieved? | Study Type           | Primary Focus                                                | System Components Addressed                                                                               |
|--------------------------------------------------|----------------------|----------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Nawadita Parajuli et al., 2019                   | Yes                  | Review [6]           | EMG pattern recognition control                              | Processing and control aspects only; no complete workflow components [6]                                  |
| P. Batista et al., 2025                          | Yes                  | Narrative review [7] | Neuromuscular robotic prostheses                             | 3D printing, AI control; no explicit parametric CAD or anthropometric integration [7]                     |
| Moosa S. M. Al Kharusi et al., 2024              | No                   | Primary study [8]    | AI-driven customized prosthetics with additive manufacturing | High-resolution imaging, AI optimization, 3D printing; no CAD models or clinician oversight mentioned [8] |
| Risana N A et al., 2025                          | No                   | Primary study [9]    | AI-powered prosthetic arm with sEMG/IMU                      | AI personalization, sEMG/IMU data; no CAD models, clinician oversight, or 3D printing mentioned [9]       |
| Dana Terrazas-Rodas & Joanna Carrión-Pérez, 2022 | No                   | Review [10]          | AI techniques for biosignal pattern recognition              | AI techniques for pattern recognition; partial anthropometric focus [10]                                  |

The included studies represent a mix of primary research (n=6) and narrative reviews (n=4), published between 2019 and 2026. Notably, only four studies had full text available beyond the abstract [2–7]. No single study addressed all five components of the complete AI-integrated prosthetic customization workflow identified in the research question. The most comprehensive implementations combined 3D printing with AI control systems [1, 2, 8], while several focused exclusively on AI-based control without addressing the broader customization pipeline [5, 9].

## Artificial Intelligence Technologies and Methods

| Study                                            | AI/ML Algorithms                                                   | Model Architecture                                        | Data Inputs                                 | Processing Mode                     | Performance Metrics                                           |
|--------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------|-------------------------------------|---------------------------------------------------------------|
| Enzo Romero et al., 2025                         | Machine learning for EMG classification [1]                        | Not specified [1]                                         | EMG signals [1]                             | Real-time [1]                       | BAM-ULA score 7.4/10; 100% gross motor, 22.2% fine motor [1]  |
| Manal Al Khaldi et al., 2025                     | Deep learning, reinforcement learning [2]                          | Training stabilized at 91% over 50 epochs [2]             | Biosensor/EMG data [2]                      | Real-time gesture recognition [2]   | 91% overall accuracy [2]                                      |
| Nadia Hachoumi et al., 2026                      | Pattern recognition, regression control, deep learning [3]         | Edge AI frameworks, ARM Cortex-M series [3]               | EMG data, computer vision [3]               | Real-time Edge AI [3]               | 92-96% classification accuracy; 95.3% fusion accuracy [3]     |
| Amira J. Zaylaa et al., 2024                     | Neural networks, deep learning [4]                                 | Optimized ANN with 3 neurons in hidden layer [4]          | EMG sensor readings [4]                     | Real-time with sigmoid feedback [4] | 89-93% accuracy range [4]                                     |
| D. K. Luu et al., 2022                           | Recurrent neural network [5]                                       | CNN + RNN with GRU, batch normalization, dropout [5]      | Multichannel nerve data [5]                 | Real-time [5]                       | 85-98% accuracy; 0.81s reaction time; 6.09 bps throughput [5] |
| Nawadita Parajuli et al., 2019                   | SVM, LDA, ANN, deep learning [6]                                   | Pattern recognition and regression methods [6]            | EMG signals [6]                             | Real-time continuous control [6]    | High accuracy mentioned without specific rates [6]            |
| P. Batista et al., 2025                          | Deep learning, hybrid models, SVM, CNN, reinforcement learning [7] | Hybrid representational and discriminative approaches [7] | EMG, force sensors, motion sensors, IMU [7] | Real-time neural decoding [7]       | Over 95% classification accuracy in some systems [7]          |
| Moosa S. M. Al Kharusi et al., 2024              | AI algorithms, machine learning [8]                                | Not specified [8]                                         | High-resolution anatomical imaging [8]      | Not specified [8]                   | Not specified [8]                                             |
| Risana N A et al., 2025                          | Machine learning algorithms [9]                                    | Not specified [9]                                         | sEMG signals, IMU data [9]                  | Not specified [9]                   | Accurate gesture recognition [9]                              |
| Dana Terrazas-Rodas & Joanna Carrión-Pérez, 2022 | AI techniques for pattern recognition [10]                         | Not specified [10]                                        | Not specified [10]                          | Not specified [10]                  | Not specified [10]                                            |

AI technologies in upper-limb prosthetic systems predominantly focused on biosignal interpretation rather than design customization. Deep learning approaches demonstrated the highest accuracy rates, with pattern recognition and regression-based control achieving 92% or greater success rates [3], substantially improved from legacy systems at 75% [3]. The most sophisticated implementations employed recurrent neural networks with convolutional layers for real-time motor decoding, achieving 97-98% accuracy for individual finger and wrist movements [5]. Edge AI architectures using ARM Cortex-M7 processors enabled control latencies below 125ms, below the human perception threshold [3]. However, only two studies explicitly described AI algorithms optimizing prosthetic designs based on anatomical data [8, 8], with most AI applications limited to motor control rather than customization workflows.

## Anthropometric Integration and CAD Systems

| Study                               | Anthropometric Capture Methods      | AI Processing of Anthropometric Data       | CAD Software/Platforms | Parametric Modeling Approach                  | Integration with Manufacturing                   |
|-------------------------------------|-------------------------------------|--------------------------------------------|------------------------|-----------------------------------------------|--------------------------------------------------|
| Enzo Romero et al., 2025            | 3D scanning [1]                     | Not explicitly mentioned [1]               | Not specified [1]      | Parametric modeling for personalization [1]   | Additive manufacturing integration [1]           |
| Manal Al Khaldi et al., 2025        | Not mentioned [2]                   | Not mentioned [2]                          | Fusion 360 [2]         | Optimized for usability and manufacturing [2] | CAD to STL conversion for 3D printing [2]        |
| Nadia Hachoumi et al., 2026         | Not mentioned [3]                   | Not mentioned [3]                          | Not mentioned [3]      | Not mentioned [3]                             | 3D printing with soft materials mentioned [3]    |
| Amira J. Zaylaa et al., 2024        | EMG sensors for muscle activity [4] | AI processes EMG for bionic control [4]    | AutoCAD [4]            | Not explicitly mentioned [4]                  | Dynamic model integration [4]                    |
| D. K. Luu et al., 2022              | Not mentioned [5]                   | Not mentioned [5]                          | Not mentioned [5]      | Not mentioned [5]                             | Not mentioned [5]                                |
| Nawadita Parajuli et al., 2019      | Not mentioned [6]                   | Not mentioned [6]                          | Not mentioned [6]      | Not mentioned [6]                             | Not mentioned [6]                                |
| P. Batista et al., 2025             | CT scans for bone assessment [7]    | AI processes nerve signals for control [7] | Not mentioned [7]      | Not mentioned [7]                             | Additive manufacturing mentioned [7]             |
| Moosa S. M. Al Kharusi et al., 2024 | High-resolution imaging [8]         | AI algorithms optimize designs [8]         | Not specified [8]      | Not explicitly mentioned [8]                  | HDPE with nano-HA via additive manufacturing [8] |
| Risana N A et al., 2025             | sEMG and IMU data [9]               | AI personalizes to user patterns [9]       | Not mentioned [9]      | Not mentioned [9]                             | Not mentioned [9]                                |



| Study                                            | Anthropometric Capture Methods       | AI Processing of Anthropometric Data       | CAD Software/Platforms | Parametric Modeling Approach | Integration with Manufacturing |
|--------------------------------------------------|--------------------------------------|--------------------------------------------|------------------------|------------------------------|--------------------------------|
| Dana Terrazas-Rodas & Joanna Carrión-Pérez, 2022 | Patient characteristics implied [10] | AI techniques for pattern recognition [10] | Not mentioned [10]     | Not mentioned [10]           | Not mentioned [10]             |

The integration of anthropometric data with AI-driven design systems remained largely underdeveloped across the included studies. Only three studies explicitly described methods for capturing anatomical data: 3D scanning for parametric modeling [1], high-resolution imaging for design optimization [8], and CT scans for preoperative planning [7]. The pathway from anthropometric capture to AI-processed parametric CAD models was minimally documented. While Romero et al. described using 3D scanning with parametric modeling for anatomical personalization [1], and Al Kharusi et al. mentioned AI algorithms processing imaging data to optimize designs [8], neither study detailed how AI suggested or modified design parameters within CAD environments. The most common CAD platforms identified were Fusion 360 [2] and AutoCAD [4], but their integration with AI systems for parameter suggestion was not specified. This disconnect between anthropometric input and AI-guided parametric design represents a significant gap in the current implementation of integrated customization workflows.

### 3D Printing Workflows and Materials

| Study                          | Materials Used                             | Print Parameter Optimization                             | Quality Control Measures                     | Design-to-Manufacturing Integration            |
|--------------------------------|--------------------------------------------|----------------------------------------------------------|----------------------------------------------|------------------------------------------------|
| Enzo Romero et al., 2025       | Not specified [1]                          | Not mentioned [1]                                        | Not mentioned [1]                            | Additive manufacturing for personalization [1] |
| Manal Al Khaldi et al., 2025   | Not specified [2]                          | Optimized in-fill and speed for structural integrity [2] | Assembly with fasteners and components [2]   | CAD to STL conversion via slicing software [2] |
| Nadia Hachoumi et al., 2026    | Flexible polymers, Shape Memory Alloys [3] | Not mentioned [3]                                        | Not mentioned [3]                            | 3D printing with soft materials [3]            |
| Amira J. Zaylaa et al., 2024   | Plastic material [4]                       | Implied through durability testing [4]                   | Testing for durability and functionality [4] | Iterative design, testing, refinement [4]      |
| D. K. Luu et al., 2022         | Not mentioned [5]                          | Not mentioned [5]                                        | Not mentioned [5]                            | Not mentioned [5]                              |
| Nawadita Parajuli et al., 2019 | Not mentioned [6]                          | Not mentioned [6]                                        | Not mentioned [6]                            | Not mentioned [6]                              |
| P. Batista et al., 2025        | Not specified [7]                          | Not mentioned [7]                                        | Not mentioned [7]                            | Modular, open-source frameworks [7]            |

| Study                                            | Materials Used                                         | Print Parameter Optimization | Quality Control Measures                              | Design-to-Manufacturing Integration                   |
|--------------------------------------------------|--------------------------------------------------------|------------------------------|-------------------------------------------------------|-------------------------------------------------------|
| Moosa S. M. Al Kharusi et al., 2024              | HDPE reinforced with micro and nano-hydroxyapatite [8] | Not mentioned [8]            | ASME standards mechanical testing; FEM validation [8] | AI algorithms optimize for additive manufacturing [8] |
| Risana N A et al., 2025                          | Not mentioned [9]                                      | Not mentioned [9]            | Not mentioned [9]                                     | Not mentioned [9]                                     |
| Dana Terrazas-Rodas & Joanna Carrión-Pérez, 2022 | Not mentioned [10]                                     | Not mentioned [10]           | Not mentioned [10]                                    | Not mentioned [10]                                    |

3D printing integration varied substantially in technical sophistication. The most advanced material system employed high-density polyethylene reinforced with micro and nano-hydroxyapatite, which underwent rigorous mechanical testing according to ASME standards and validation via finite element modeling [8]. This study demonstrated that nano-HA inclusion significantly improved tensile strength and modulus for biomechanical support [8]. Other implementations used standard plastic materials [4] or were exploring soft materials including flexible polymers and shape memory alloys for more biologically compatible designs [3]. Print parameter optimization was explicitly described in only one study, where in-fill and speed parameters were balanced for structural integrity and surface finish [2]. The translation from AI-generated designs to printable models typically involved CAD-to-STL conversion through slicing software [2], but automated integration between AI design optimization and manufacturing constraints was not demonstrated in any study. Quality control approaches ranged from assembly validation [2] to comprehensive durability and functionality testing [4], with the most rigorous implementing standardized mechanical testing protocols [8].

### Clinician Oversight Mechanisms

None of the ten included studies provided information about clinician roles, oversight mechanisms, or integration points within AI-enhanced upper-limb prosthetic customization systems [1–10]. No studies described how clinicians review, modify, or approve AI-generated suggestions, nor did any detail user interfaces or tools designed for clinician interaction with AI systems. The balance between AI automation and clinician control, mechanisms for clinician feedback integration, and specific involvement points throughout the customization workflow were entirely absent from the literature. This represents a critical gap, as clinical oversight is typically essential in medical device customization to ensure safety, appropriateness, and individualized optimization beyond automated algorithms.

### Functional Performance and Validation

| Study                                            | Functionality Metrics                   | Accuracy/Success Rates                 | Usability Outcomes                                 | Cost and Time Efficiency                                  | User/Clinical Acceptance                          |
|--------------------------------------------------|-----------------------------------------|----------------------------------------|----------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------|
| Enzo Romero et al., 2025                         | BAM-ULA score 7.4/10 (SD 0.7) [1]       | 100% gross motor; 22.2% fine motor [1] | Robust performance [1]                             | Affordable solution via AI and additive manufacturing [1] | Comparable to high-end commercial devices [1]     |
| Manal Al Khaldi et al., 2025                     | 91% AI accuracy [2]                     | Not specified [2]                      | High accuracy for gripping, pinching, twisting [2] | 81% energy efficiency; 15% power reduction [2]            | Improved over previous models [2]                 |
| Nadia Hachoumi et al., 2026                      | Control latency <125ms [3]              | 92%+ success rates with AI [3]         | 30% reduction in cognitive load [3]                | 15.7% battery longevity increase [3]                      | Improved autonomy and function [3]                |
| Amira J. Zaylaa et al., 2024                     | Strengthen grip, high functionality [4] | 89-93% AI control accuracy [4]         | Lightweight with no delay [4]                      | Most cost-effective among compared designs [4]            | Comparable to i-Limb and Bebionic [4]             |
| D. K. Luu et al., 2022                           | 6 DOF control [5]                       | 97-98% accuracy for movements [5]      | Intuitive control via motor intent [5]             | Not mentioned [5]                                         | Consistent performance over 16 months [5]         |
| Nawadita Parajuli et al., 2019                   | Multiple DOF limitations [6]            | Not specified [6]                      | Socket interference, discomfort noted [6]          | Need for low-cost materials [6]                           | Need for better comfort and reliability [6]       |
| P. Batista et al., 2025                          | Improved range of motion [7]            | Not specified [7]                      | Enhanced comfort with osseointegration [7]         | Not mentioned [7]                                         | Significant user satisfaction with PR control [7] |
| Moosa S. M. Al Kharusi et al., 2024              | Improved tensile strength [8]           | Not specified [8]                      | Enhanced comfort and functionality [8]             | Not mentioned [8]                                         | Positive user feedback [8]                        |
| Risana N A et al., 2025                          | 7 DOF control [9]                       | Accurate gesture recognition [9]       | Lightweight, ergonomic design [9]                  | Not mentioned [9]                                         | Improved usability via AI personalization [9]     |
| Dana Terrazas-Rodas & Joanna Carrión-Pérez, 2022 | Not specified [10]                      | Not specified [10]                     | Not specified [10]                                 | Not mentioned [10]                                        | Not specified [10]                                |

Functional performance outcomes demonstrated substantial variability in both measurement approaches and re-

ported success. Control accuracy for AI-based systems ranged from 85-98% [5] across studies, with the highest performance achieved through peripheral nerve interfaces using recurrent neural networks [5]. Pattern recognition approaches showed median reaction times of 0.81 seconds with information throughput of 6.09 bits per second at 99% accuracy [5]. Standardized functional assessments were rare; only one study employed the validated BAM-ULA protocol, achieving a mean score of 7.4 out of 10 [1]. Task-specific success rates revealed a consistent pattern: gross motor tasks achieved 100% success while fine motor tasks succeeded in only 22.2% of trials [1], highlighting the persistent challenge of dexterous manipulation. Energy efficiency represented a key advancement, with optimized systems demonstrating 81% energy conversion rates and 15% power consumption reduction compared to previous designs [2]. Edge AI implementations reduced control latencies below the human perception threshold of 125ms [3], while integration of computer vision with EMG data decreased user cognitive load and compensatory shoulder movements by 30% [3]. Long-term stability was documented in one study showing consistent signal strength over 16 months with improving reaction times [5].

Cost effectiveness was explicitly addressed in three studies, all reporting affordability improvements through integration of AI and additive manufacturing [1, 2, 4]. However, none provided quantitative cost comparisons or time reduction metrics relative to traditional customization methods. User satisfaction data was limited and primarily qualitative, with reports of enhanced comfort and functionality [8] and favorable comparisons to commercial devices like i-Limb and Bebionic [4]. Clinical acceptance was implied through performance comparability to high-end commercial systems [1] and significant satisfaction gains with pattern recognition control [7], though no studies presented systematic clinical validation or regulatory approval pathways.

## **Synthesis: Integration Gaps and Implementation Patterns**

The literature reveals a fundamental disconnect between the conceptual vision of AI-integrated prosthetic customization systems and current implementation realities. While the research question sought systems combining anthropometric input, parametric CAD models, AI parameter suggestion, clinician oversight, and 3D printing workflows, no study demonstrated integration across all five components. This absence reflects not methodological inconsistency but rather distinct research trajectories addressing different elements of the prosthetic development pipeline.

### **Component-Specific vs. Holistic Integration**

Studies clustered into three distinct implementation approaches. First, control-focused systems (6 of 10 studies) concentrated exclusively on AI-enhanced motor control through biosignal interpretation [3–6, 9, 10], achieving impressive accuracy rates (85-98%) [5] but addressing none of the customization workflow components. Second, manufacturing-oriented implementations (3 studies) combined 3D printing with varying levels of AI integration for design or control [1, 2, 8], incorporating 2-3 workflow components but omitting clinician oversight entirely. Third, review papers examined technological capabilities without demonstrating integrated systems [3, 6, 7, 10].

This pattern suggests that AI integration into prosthetic customization workflows remains in an early stage where component-level optimization precedes systems-level integration. The most advanced implementations by Romero et al. combined 3D scanning, parametric modeling, AI-driven control, and additive manufacturing [1], yet even this system lacked explicit mechanisms for AI-based parameter suggestion within the CAD environment or clinician oversight protocols. The gap between anthropometric capture and AI-processed design modification represents a particularly underdeveloped transition point: while studies captured anatomical data via 3D scanning [1] or high-resolution imaging [8], none detailed algorithms that translated this data into AI-suggested parametric adjustments within CAD systems.

### **Accuracy-Cost Trade-offs in AI Implementation**

The most sophisticated AI architectures achieved superior control accuracy but at implementation costs that may limit accessibility. Systems employing recurrent neural networks with convolutional layers for peripheral nerve interfaces demonstrated 97-98% accuracy for individual finger and wrist movements [5], substantially outperforming pattern recognition approaches at 92-96% [3]. However, these advanced systems required invasive neural interfaces and specialized hardware, while simpler EMG-based systems achieved 89-93% accuracy [4] using non-invasive sensors compatible with 3D-printed prosthetics explicitly designed for affordability [1, 4].

This accuracy-accessibility trade-off may explain the bifurcation in research approaches. Studies prioritizing accessibility focused on integrating low-cost 3D printing with simpler AI control systems [1, 2, 4], accepting slightly lower accuracy in exchange for reduced barriers to implementation. Conversely, studies pursuing maximum performance implemented sophisticated AI architectures without addressing customization workflows or manufacturing integration [5]. The optimal balance likely depends on user context: populations in low-resource settings may benefit more from 91% accurate systems that cost substantially less [2, 4], while users requiring high dexterity for professional tasks may justify the investment in 97-98% accurate but more expensive systems [5].

### **The Absence of Clinician Oversight as Systematic Gap**

The complete absence of clinician oversight mechanisms across all ten studies [1–10] represents a critical gap that cannot be explained by study heterogeneity or methodological variation. This universal omission suggests that AI-integrated prosthetic customization research has proceeded without adequate consideration of clinical decision-making roles. Medical device customization typically requires clinical expertise to evaluate appropriateness, safety, and patient-specific contraindications beyond algorithmic optimization. The absence of described interfaces for clinician review of AI-generated suggestions, mechanisms for manual override, or protocols for clinical validation indicates these systems remain research prototypes rather than clinically deployable solutions.

This gap becomes particularly concerning when considered alongside the performance limitations evident in functional testing. While gross motor tasks achieved 100% success, fine motor tasks succeeded in only 22.2% of trials [1], suggesting AI systems cannot yet reliably optimize prosthetics for all functional requirements. Clinical oversight would provide essential judgment about acceptable performance trade-offs, task prioritization based on user goals, and safety considerations when automated systems encounter edge cases. Future implementations must integrate clinical decision points throughout the workflow: reviewing AI-processed anthropometric data for anatomical anomalies, approving or modifying AI-suggested design parameters, validating mechanical safety before fabrication, and assessing functional performance post-fitting.

### **Material Science as Enabling Factor**

The integration of advanced materials in 3D printing workflows represents an underexplored enabler of AI customization systems. The single study employing high-density polyethylene reinforced with nano-hydroxyapatite achieved significant improvements in tensile strength and modulus while maintaining compatibility with additive manufacturing [8]. This material advancement enabled AI-optimized designs validated through finite element modeling [8], demonstrating that sophisticated material systems can support the mechanical demands of personalized geometries generated by AI algorithms. However, most studies used standard plastics [4] or did not specify materials [1], suggesting this integration opportunity remains largely unexploited. Future systems combining AI-driven topological optimization with multi-material 3D printing could generate designs impossible through traditional manufacturing, personalized to individual biomechanical loading patterns while meeting strength requirements validated through computational modeling.

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